Right Triangles & Trigonometry Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Pythagorean theorem

ORIGINAL PROBLEM

A right triangle has legs 6 and 8. Find the hypotenuse.

INCORRECT WORKED SOLUTION

$c^2 = 6^2 + 8^2$

$c^2 = 100$

$c = 50$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Finding a leg

ORIGINAL PROBLEM

A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

INCORRECT WORKED SOLUTION

$a^2 = 13^2 + 5^2$

$a^2 = 194$

$a \approx 13.9$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Converse of the theorem

ORIGINAL PROBLEM

Is a triangle with sides 8, 15 and 17 a right triangle?

INCORRECT WORKED SOLUTION

$8 + 15 = 23$

$23 \neq 17$

It is not a right triangle.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Trig ratio

ORIGINAL PROBLEM

In a right triangle, the side opposite ∠A is 5, the adjacent side is 12 and the hypotenuse is 13. Find sin A.

INCORRECT WORKED SOLUTION

$$\sin A = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\sin A = \frac{12}{13}$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Solving for a side

ORIGINAL PROBLEM

$\angle A = 35°$. The side adjacent to $\angle A$ is 20. Find the opposite side x.

INCORRECT WORKED SOLUTION

$$\tan 35° = \frac{20}{x}$$

$$x = \frac{20}{\tan 35°}$$

$$x \approx 28.6$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Finding an angle

ORIGINAL PROBLEM

In a right triangle, the side opposite $\angle\theta$ is 7 and the hypotenuse is 10. Find θ.

INCORRECT WORKED SOLUTION

$$\cos \theta = \frac{7}{10} = 0.7$$

$$\theta = \cos^{-1}(0.7)$$

$$\theta \approx 45.6°$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Special right triangle

ORIGINAL PROBLEM

A 45°-45°-90° triangle has legs of length 6. Find the hypotenuse.

INCORRECT WORKED SOLUTION

hypotenuse = leg $\cdot \sqrt{3}$

hypotenuse = $6\sqrt{3}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Special right triangle

ORIGINAL PROBLEM

In a 30°-60°-90° triangle the shortest leg is 5. Find the hypotenuse.

INCORRECT WORKED SOLUTION

hypotenuse = short leg $\cdot \sqrt{3}$

hypotenuse = $5\sqrt{3}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Ladder problem

ORIGINAL PROBLEM

A 15-ft ladder leans on a wall with its base 5 ft from the wall. How high up the wall does it reach?

INCORRECT WORKED SOLUTION

$h^2 = 15^2 + 5^2$

$h^2 = 250$

$h \approx 15.8$ ft

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Angle of depression

ORIGINAL PROBLEM

From the top of a 50-ft tower, the angle of depression to a boat is 25°. How far is the boat from the base of the tower?

INCORRECT WORKED SOLUTION

$\tan 25° = \frac{d}{50}$

$d = 50 \cdot \tan 25°$

$d \approx 23.3$ ft

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points